

HTML Forms: Collecting User Data

A beginner's guide to building interactive, data-collecting web pages with HTML forms.

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What Are HTML Forms?

Forms are the primary way websites collect information from users and send it to a web server. They're everywhere — and mastering them is essential for any web developer.

🔐 Login

Username & password forms

📝 Registration

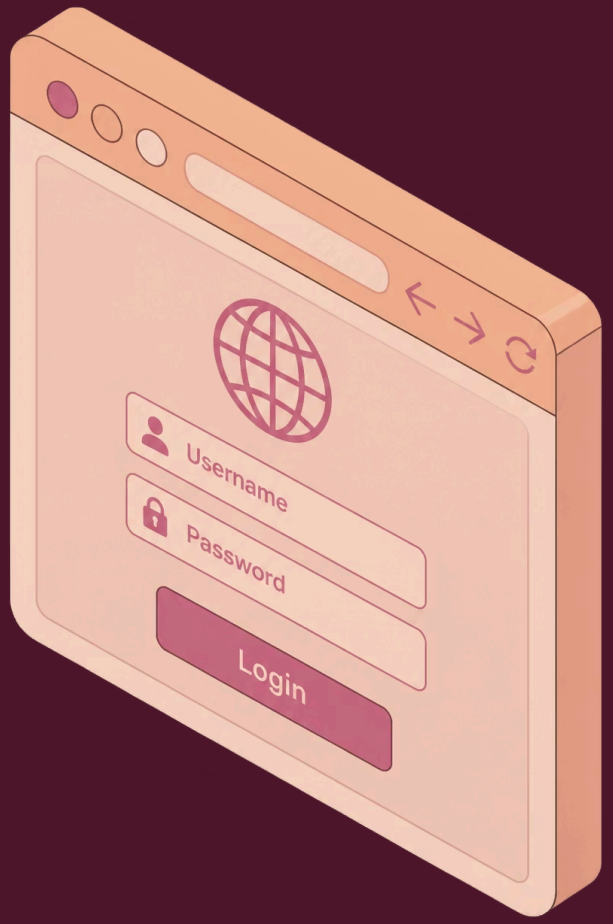
Sign-up and account creation

🔍 Search

Query input & filters

🛒 Checkout

Payments & order details



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Why Do We Use Forms?



Forms power the web's interactivity

→ Collect User Information

Names, emails, preferences, and profile data

→ Authenticate Users

Login and password verification to protect accounts

→ Process Transactions

Online shopping, payments, and order management

→ Gather Feedback

Surveys, ratings, and comments from users

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The <form> Tag: Your Container

Every form starts with the <form> element. Three key attributes control how your form behaves:

action

Specifies **where** the form data is sent after submission.

Example: action="register.php"

method

Controls **how** data is sent to the server.

- **GET** — visible in URL, less secure
- **POST** — hidden, more secure ✅

autocomplete

Lets the browser remember previous entries.

Values: on or off

```
<form action="register.php" method="POST" autocomplete="on">
 <!-- form fields go here -->
</form>
```

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Input Types: Text, Email, Password & Number

The most common input types

The `<input>` element is the workhorse of HTML forms. Change its `type` attribute to unlock different behaviors and validations automatically.

```
<!-- Text input -->
<label>Name:</label>
<input type="text" name="name">

<!-- Email input -->
<label>Email:</label>
<input type="email" name="email">

<!-- Password input -->
<label>Password:</label>
<input type="password" name="pass">

<!-- Number input -->
<label>Age:</label>
<input type="number" name="age">
```

text

Single-line text — names, cities

email

Auto-validates email format

password

Hides characters as you type

number

Accepts only numeric values

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More Input Types: Date, File, Color & Range



type="date"

Renders a calendar date picker — great for birthdays, deadlines, or booking forms.



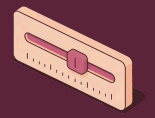
type="file"

Opens a file browser so users can upload documents, images, or other files.



type="color"

Displays a color picker tool — useful for customization or preference settings.



type="range"

A draggable slider for selecting values within a defined min and max range.

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Checkboxes & Radio Buttons

Checkboxes — Multiple Selections ✅

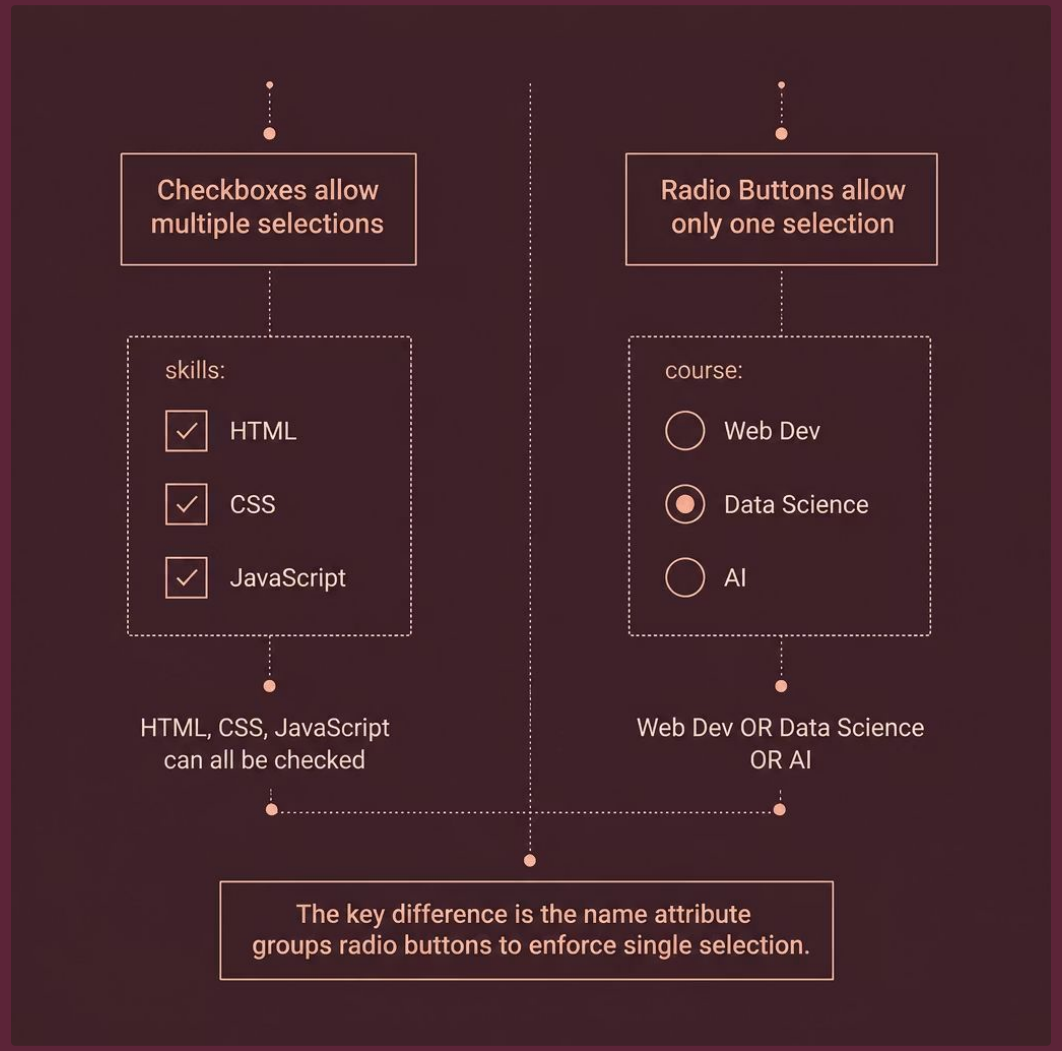
Use checkboxes when users can pick **more than one** option at a time — like skills or interests.

```
<input type="checkbox" name="skills" value="html"> HTML
<input type="checkbox" name="skills" value="css"> CSS
<input type="checkbox" name="skills" value="js"> JavaScript
```

Radio Buttons — One Selection Only 🔘

Use radio buttons when users must pick **exactly one** option. Group them with the same `name` attribute.

```
<input type="radio" name="course" value="web"> Web Dev
<input type="radio" name="course" value="data"> Data Science
<input type="radio" name="course" value="ai"> AI
```



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Dropdowns & Text Areas

`<select>` — Dropdown Menus

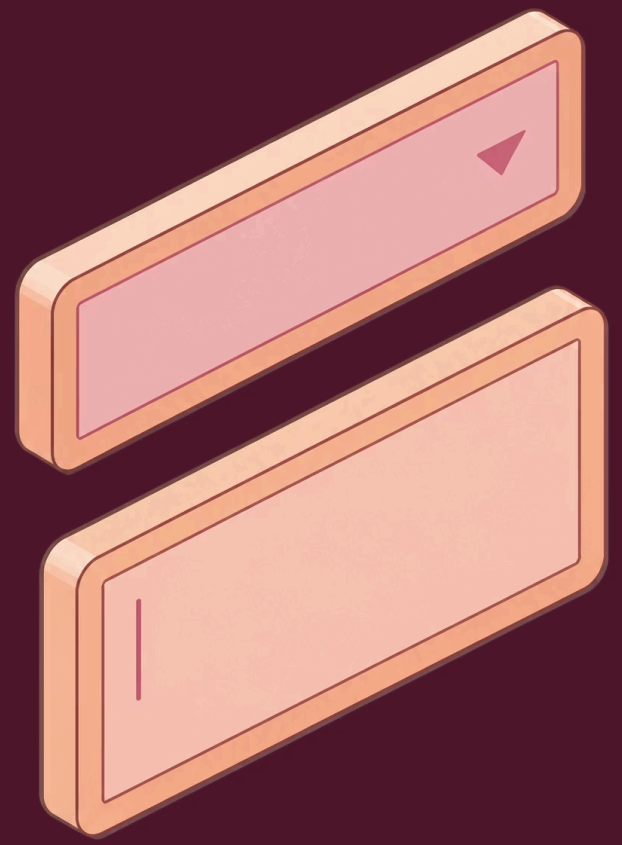
Create a compact dropdown with `<option>` items inside. Use `<optgroup>` to group related choices.

```
<select name="language">
  <optgroup label="Web">
    <option>HTML</option>
    <option>CSS</option>
  </optgroup>
  <optgroup
label="Programming">
    <option>Python</option>
    <option>JavaScript</option>
  </optgroup>
</select>
```

`<textarea>` — Large Text Input

Use `<textarea>` for longer responses like addresses, feedback, or comments. Set `rows` and `cols` to control size.

```
<textarea name="bio"
 rows="5"
 cols="40"
 placeholder="Tell us about
yourself...">
</textarea>
```



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Built-in Validation (HTML5)

HTML5 gives you powerful validation attributes — no JavaScript required! Add them directly to your `<input>` tags.

1

required

The field **must** be filled before the form can be submitted.

```
<input type="text" required>
```

2

minlength / maxlength

Controls the minimum and maximum number of characters allowed.

```
<input type="text" minlength="3" maxlength="20">
```

3

min / max

Sets number or date range boundaries for valid input values.

```
<input type="number" min="18" max="99">
```

4

pattern

Matches input against a specific format using a regex expression.

```
<input type="tel" pattern="[0-9]{10}">
```

Validation protects your form from bad data — always validate on the server side too for real security.

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Complete Student Registration Form

Your lab task is to **build and test** a complete registration form that combines everything you've learned. Your form must include all of the following elements:

📋 Text Inputs

Name, email, and password fields

📅 Date Picker

Date of birth selection

🔘 Radio Buttons

Select your course program

☑️ Checkboxes

Choose your skills/interests

📂 File Upload

Upload a profile photo

✅ Validation

Required fields, length limits & patterns

Don't forget your `Submit` and `Reset` buttons! Test your form in the browser and make sure validation works before marking complete.

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